

JOEL HÄGVALL

Software Engineer

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PROFILE

Software Engineer and Founding Engineer building AI products end-to-end, with hands-on experience in RAG, AI agents, eval-oriented testing, observability and product workflows for real users. Currently working at Scania while serving as the sole engineer in a three-person founding team at ResiliaAI.

EXPERIENCE

Software Developer

Scania | Sep 2024 - Present

- Owned migration and development of internal .NET/C# microservices, from planning through implementation.
- Built map-based internal web functionality using TypeScript and React.
- Worked with AWS and cloud-related development tasks across internal tooling.
- Developed enterprise service management solutions and internal tooling across business processes.
- Built workflows, automations and integrations in BMC Remedy, including low-code configuration and process logic.

Founding Engineer

ResiliaAI | 2025 - Present

AI-powered resilience and crisis preparedness platform for housing associations and organizations.

- Founding Engineer and sole software engineer in a three-person founding team, building the platform end-to-end alongside a full-time role at Scania.
- Built and deployed product functionality end-to-end using Vite, React, TypeScript, Node.js 22, Express.js, PostgreSQL, Supabase, Neon, Railway and Vercel.
- Developed AI-assisted product workflows including RAG over 24,000+ vectorized Q&A entries, embeddings, vector search, AI agents and multi-provider LLM orchestration.
- Built automated testing workflows with Jest and Playwright across unit, integration and end-to-end testing, created alongside feature development.
- Integrated test automation into CI/CD with sharding to support reliable feature delivery as the codebase grows.
- Designed evaluation and observability workflows for AI features, including prompt and agent iteration, failure-mode analysis, regression checks and logged model behavior across RAG workflows, supporting EU AI Act-oriented compliance work.

R&D Summer Intern

Ericsson | Summer 2023 & Summer 2024

- Developed Python tooling for radio performance verification teams.
- Improved Java code quality by resolving SonarQube and SonarLint findings, bug fixes and maintainability issues across internal software.
- Used Jenkins as part of development and testing workflows.
- Collaborated in an R&D engineering environment focused on automation, reliability and technical validation.
- Returned for a second summer after receiving a return offer.

Vice Head of Marketing

Systemvetardagen | Sep 2023 - Mar 2024

- Led marketing initiatives for a technology career fair with 3,000+ attendees and managed partnerships, planning and team coordination.

PROJECTS

Data Wipe Mailer

Built and launched a privacy-focused GDPR deletion tool with Next.js, TypeScript, Tailwind CSS and shadcn/ui, hosted on Vercel. The project generated 200,000+ LinkedIn views and 2,000+ likes after launch.

J.A.R.V.I.S

Private local AI assistant for macOS built with SwiftUI, Ollama and MCP servers for personal notes, tools and local-first AI workflows.

Tor Onion Site Scraper

Python-based crawler and analysis platform for a university thesis, retrieving Tor HTML pages, extracting marketplace data and supporting quantitative and qualitative analysis.

Carspotter

Student-built social platform for car enthusiasts using Flutter, Firebase, Google Maps API and Plate Recognizer.

TECHNICAL SKILLS

Frontend: React, Vite, Next.js, TypeScript, JavaScript, TanStack Query, TanStack Table, React Hook Form, Tailwind CSS, Radix UI, shadcn/ui

Backend: Node.js 22, Express.js, .NET, C#, Java, PostgreSQL, Supabase, Neon, REST APIs, authentication and authorization

AI & Automation: Vercel AI SDK, AI Agents, RAG, Embeddings, Vector Search, MCP, OpenAI, Anthropic Claude, Google Gemini, AI-assisted development

Testing & CI/CD: Jest, Playwright, unit/integration/E2E workflows, test sharding, CI/CD, Jenkins, OpenAPI checks

Cloud & Infrastructure: AWS, Docker, Railway, Vercel, Azure Key Vault, Git, GitHub, logging and observability workflows

EDUCATION

B.Sc. Computer and Systems Sciences

Stockholm University | 2021 - 2024